

## QUESTION -

### 1358. Number of Substrings Containing All Three Characters

Medium

Topics

Companies

Hint

Given a string `s` consisting only of characters *a*, *b* and *c*.

Return the number of substrings containing **at least** one occurrence of all these characters *a*, *b* and *c*.

#### Example 1:

**Input:** `s = "abcabc"`

**Output:** 10

**Explanation:** The substrings containing at least one occurrence of the characters *a*, *b* and *c* are `"abc"`, `"abca"`, `"abcab"`, `"abcabc"`, `"bca"`, `"bcab"`, `"bcabc"`, `"cab"`, `"cabc"` and `"abc"` (**again**).

#### Example 2:

**Input:** `s = "aaacb"`

**Output:** 3

**Explanation:** The substrings containing at least one occurrence of the characters *a*, *b* and *c* are `"aaacb"`, `"aacb"` and `"acb"`.

#### Example 3:

**Input:** `s = "abc"`

**Output:** 1

#### Constraints:

- $3 \leq s.length \leq 5 \times 10^4$
- `s` only consists of `'a'`, `'b'` or `'c'` characters.

SOLUTION -

```
class Solution:
    def numberOfSubstrings(self, s: str) -> int:

        left = 0
        answer = 0

        # Frequency of a, b and c
        count = {'a': 0, 'b': 0, 'c': 0}

        # Expand the window
        for right in range(len(s)):

            count[s[right]] += 1

            # While window contains all three characters
            while count['a'] > 0 and count['b'] > 0 and count['c'] > 0:

                # Every extension of this window is also valid
                answer += len(s) - right

                # Shrink the window
                count[s[left]] -= 1
                left += 1

        return answer
```